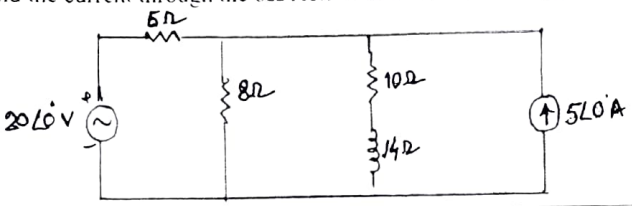
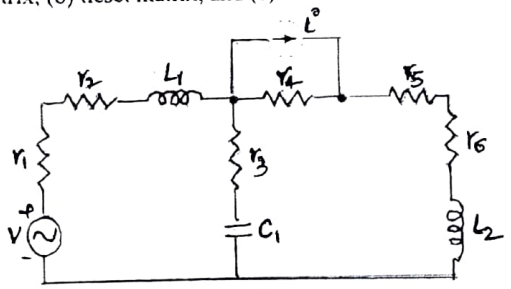
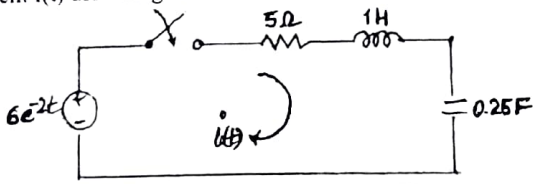
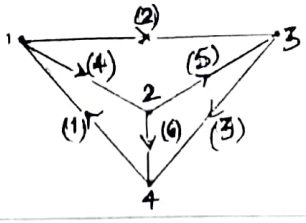
MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Fourth Semester, March, 2025

Paper Code: EEC-206 | Subject: Network Analysis and Synthesis
Time: 1½ Hrs. | Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Questions	Max. Marks	CO(s)
1	(a) State Norton's Theorem.	3	1
	(b) Define twigs and links of tree.	2	1
	(c) Find the Laplace transform of e^{t-a}, $t > a$	3	2
	(d) Define transient and steady state Response.	2	2
2	(a) Find the current through the 8Ω resistor in the network of figure. 	5	1
	(b) For the circuit shown in figure, draw the oriented graph and write (a) incidence matrix, (b) tieset matrix, and (c) cutset matrix 	5	1
3	(a) Find the Laplace transform of $\int_0^t e^{-2t} t^3 \, dt$	5	2
	(b) For the network shown in figure, the switch is closed at t = 0. Determine the current i(t) assuming zero initial conditions in the network elements. 	5	2
4	The graph of a network is shown in figure. Write the (a) incidence matrix, (b) tieset matrix, and (c) f-cutset matrix. 	10	2

(Please write your Enrolment No. immediately) Enrolment No. _____

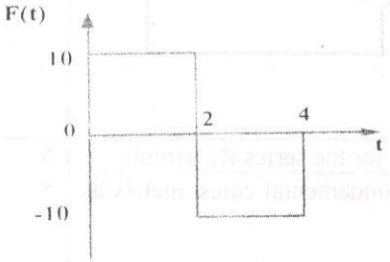
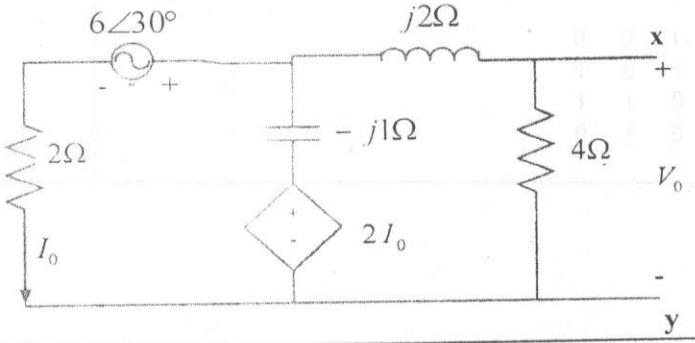
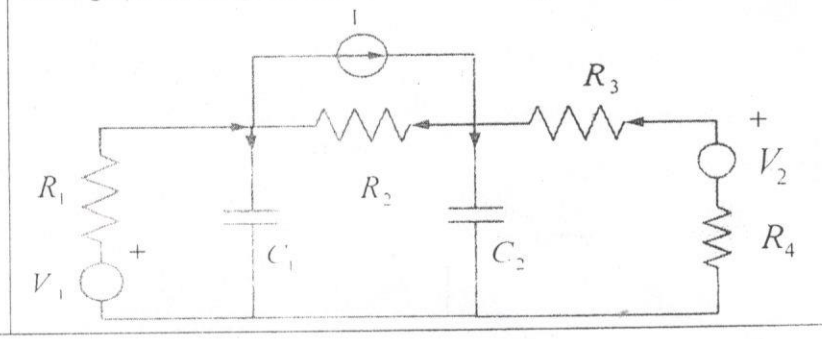
MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

4th Semester, April, 2024

Paper Code: EEC206 Subject: Network Analysis and Systems
Time: 1½Hrs. Max. Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Question	Max. Marks	CO(s)
1. (a)	Define graph and incidence matrix.	2	
(b)	Find the laplace transform of $F(t)= \sin\omega t$.	2	
(c)	For the current $i(t)= 5u(t)-3e^{-2t}$. Find I(s).	2	
(d)	Sketch the signal from the given expression $U(t-6)+U(t-4)$	2	
(e)	Find the expression for i(t) in the form of standard signals 	2	
2. (a)	Obtain the output voltage V_0 across x-y using mesh analysis in the network of the figure below 	7	
(b)	Draw graph for the given circuit as shown in figure and find A. 	3	

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Enrolment No. _____

MID TERM EXAMINATION

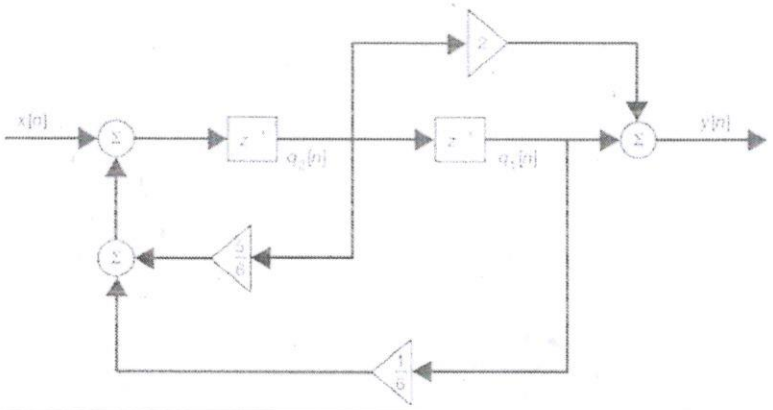
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

CSE/IT- 4th Semester, April, 2024

Paper Code: EEC-208 **Subject: Circuits and Systems**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Check linearity and time-invariance of system y(n) = 2n+x(n).	2	CO1
1 (b)	Explain an overdamped response of series RLC circuit and sketch the curve.	2	CO2
1 (c)	Find the Laplace transform and associated ROC of signal $x(t) = e^{-2t}$ [u(t) - u(t - 5)].	3	CO1
1 (d)	For the LTI system described by the following equation. Determine the impulse function of system. $y''(t) + y'(t) - 2y(t) = x(t)$	3	CO2
2 (a)	Given $X(z) = \frac{z(z-4)}{(z-1)(z-2)(z-3)}$ (i) State all the possible regions of convergence. (ii) For which ROC, z-transform X(z) is causal sequence	6	CO1
2 (b)	Find the state space representation of the discrete-time system with the state variables $q_1[n]$ and $q_2[n]$ shown in Figure. 	4	CO1
3 (a)	Obtain expression for current i(t) and voltage across the inductor in the circuit shown in figure.	4	CO2

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